



Full Length Article

Seismic performance evaluation of isolated liquid storage tanks with passive variable damping and stiffness devices

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ABSTRACT

Liquid storage tank (LST), as critical infrastructure in the petrochemical industry, is highly vulnerable to earthquake-induced damage, which may trigger secondary hazards such as fires and explosions. Seismic isolation has been widely recognized as an effective strategy to mitigate seismic impacts on LSTs. To achieve effective vibration control while preventing isolation system failure due to excessive displacement, a novel passive variable damping and stiffness device (PVDSD) was proposed. Performance testing verifies that the PVDSD delivers real-time variable damping forces responsive to velocity and provides stepwise stiffness activation upon excessive displacement within the isolation layer. Subsequently, the numerical model of PVDSD was developed and validated against test results. Integrated with lead-rubber bearings (LRBs), a new seismic isolation system was formulated and benchmarked against alternative isolation configurations under seismic excitations spanning four hazard levels. Five key engineering demand parameters (EDPs) including the isolation layer displacement, base shear, overturning moment, sloshing height and hydrodynamic force are evaluated and compared. Results demonstrate that the PVDSD isolation system exhibits a superior performance on energy dissipation under design-based earthquakes and effectively mitigates residual displacements under maximum considered and rare earthquake scenarios. Although marginal increases in base shear and sloshing height are observed, these remain significantly lower than those in non-isolated systems. In general, the PVDSD isolation system substantially enhances the seismic resilience of LSTs by providing robust energy dissipation ability at low- and middle-intensity earthquakes and improved overturning resistance at high-intensity.

1. Introduction

The petrochemical industry serves as a vital pillar of a nation's economy. Under seismic excitation, the failure of pipelines or storage tanks probably result in the release of flammable and explosive substances, potentially triggering secondary disasters such as fires and explosions [1]. Moreover, the leakage of toxic gases or liquids poses severe risks to public safety and property. As shown in Fig. 1, one of the most catastrophic tank failure incidents occurred during the 1978 Miyagi Earthquake, where spilled oil not only inundated the refinery but also polluted the adjacent sea, posing a significant public endangerment and severe environmental damage. As a result, protecting storage tanks from seismic disasters and enhancing the seismic resilience is of paramount societal importance.

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Seismic isolation systems mitigate structural responses by decoupling the natural vibration frequency of a structure from the dominant frequency components of seismic waves, effectively elongating the structural period and thereby avoiding resonance with predominant ground motion frequencies [3,4]. Compared with non-isolated storage tanks, isolated tanks achieve a 60 %–80 % reduction in peak accelerations, while simultaneously suppressing liquid sloshing and preventing long-term fatigue damage to tank walls caused by sustained hydrodynamic forces [5]. Shrimali et al. [6] conducted comprehensive comparative analysis of liquid storage tanks (LSTs) equipped with various isolation systems and demonstrated that the isolation systems are effective in attenuating the earthquake acceleration transmitted to the tank. Malhotra et al. [7] proposed the seismic base isolation of ground supported cylindrical liquid storage tanks, showing such systems markedly reduce the hydrodynamic base shears, overturning moments, and axial compressive stresses in the tank wall. Rawat et al. [8] investigated the effectiveness of an innovative Oblate Spheroid base isolation system for LSTs. Zhang et al. [9] developed a friction pendulum inerter system, which significantly enhanced the seismic performance of storage tanks. Panchal et al. [10,11] investigated the effectiveness of Variable Curvature Friction Pendulum Systems (VCFPSs) and the Variable Frequency Pendulum Isolator (VFPI) in controlling base shear, sloshing displacement, and impulsive displacement of storage tanks subjected to near-fault ground motions. Contemporary research further reveal that augmenting the damping in lead-rubber bearings (LRBs) is effective in controlling base shear and overturning moments, though at the expense of increased isolation displacements over time [12].

Although seismic isolation effectively decreases vibrations transmitted to the superstructure, a critical challenge in the design of base-isolated structures lies in accommodating large displacements in the isolation layer during seismic loading with long predominant periods [13–15]. Once the displacement exceeds the expected range, the isolation devices are at risk of damage or failure. The inclusion of energy-dissipating and displacement-limiting devices in the isolation layer can further suppress vibration amplitudes and mitigate the impact on the structure [16–18].

Several studies emphasize the necessity of incorporating supplementary dampers into the isolation layer to reduce base displacement without amplifying the response of the superstructure [19–21]. Güler et al. [22] demonstrated that adding dampers to the seismic isolation system can reduce the displacement of the seismic isolation layer. Similarly, the Architectural Institute of Japan [23] emphasized that the integration of damping devices can mitigate both shear forces and displacement response within the isolation layer. Traditional viscous dampers, widely adopted for their structural simplicity and independence from external power sources, are nonetheless constrained by their fixed damping coefficients, limiting their adaptability under variable loading conditions. Consequently, increasing research attention has been directed toward passive variable damping technologies, including displacement-dependent passive viscous dampers [24–26]. Li and Shi et al. [27,28] designed a velocity-responsive passive viscous damper with adjustable orifice size, conducting experimental validations and parametric analyses that confirmed its efficacy in attenuating wind-induced vibrations. These investigations collectively demonstrate that variable damping systems, offering a broader and more adaptable range of damping coefficients, are particularly well-suited for integration into seismic isolation layers.

Researchers have also proposed the incorporation of displacement-limiting devices within isolation systems. These devices engage when the displacement of the isolation bearing surpasses a predefined threshold, enabling internal elastic elements to collide with the support structure to constrain further movement [29–33]. However, most existing limiters are designed to be activated only exceed the permissible threshold, exhibiting a lacking adaptability under varying seismic intensities. Liquid storage tanks as important industrial structures exhibit more complicated dynamic behavior compared to other structures such as buildings and bridges due to fluid structure interaction effects [34]. Most of the research on seismic isolation and displacement limitation predominantly centers on bridges and building structures, whereas studies addressing the combined application of passive variable damping and displacement constraint mechanisms for the seismic protection of liquid storage tanks (LSTs) are notably limited in the existing literature.

The major objective of this study is to develop an isolation system incorporating a novel passive variable damping and stiffness device and evaluate its efficiency in controlling the seismic performance of LST under multiple hazard levels. The structure of this paper is organized as follows: Section 2 introduces the working principle and performance test results of the passive variable damping and stiffness device (PVDSD). Section 3 outlines the numerical models of both the LST and the PVDSD, which are also validated against



Fig. 1. Spilled oil inundated a refinery area in Sendai, Japan during the 1978 Miyagi earthquake [2].

test results. Section 4 investigates the seismic performance of LST with the developed isolation system, including structural displacement, base shear, overturning moments, sloshing wave heights and peak hydrodynamic pressures of the LST, under four seismic hazard scenarios. Finally, Section 5 summarizes the conclusions derived from this study.

2. Overview and performance test of PVDSD

2.1. The configuration and working principle of PVDSD

PVDSD comprises a variable damping device and a multi-stage variable stiffness device, designed to achieve adaptive seismic control. The variable damping device is an enhanced version based on references [27,28], and the damping force it generates increases nonlinearly in real-time with rising loading velocities. This design offers more stable performance, a simpler structural configuration, and fewer adjustable parameters. The variable stiffness device, connected in parallel with the damping unit, provides discrete stiffnesses. Both components are actuated by a through-going rod, ensuring synchronized response under simultaneous loading. A schematic representation of the PVDSD is presented in Fig. 2.

2.1.1. Variable damping device

The variable damping device (Fig. 2(a)) comprises a damping cylinder, piston rod, two damping valves and external pipelines. Each damping valve integrates a check valve and a cone valve, arranged antisymmetrically on either side of the cylinder. The check valves regulate the fluid's flow direction, enabling variable damping in both loading directions to accommodate bidirectional excitation. As illustrated in Fig. 3 the cone valve consists of a valve body, a conical valve core, a spring, and a threaded cap. Under fluid pressure, the conical valve core is driven to compress the spring, progressively reducing the conical orifice area and thereby producing the increasing variable damping effect.

A schematic illustration of the cone valve's working principle is presented in Fig. 4. As the external excitation velocity (corresponding to the flow velocity of the damping fluid), increases from v_1 to v_2 and subsequently to v_3 , the hydrodynamic force drives the valve core rightward, causing the cone valve orifice to transition from a fully open state (Stage 1), to a partially open state (Stage 2), and eventually to a nearly closed state (Stage 3). The progressive reduction in orifice area restricts fluid flow, thereby resulting in a significant increase in damping force.

In addition to the fluid flow, the spring within the cone valve plays a critical role in governing the motion of the valve core. Consequently, both the stiffness and the initial length of the spring significantly influence the variable damping characteristics. When the spring is in its free-length state (its length matches the internal cavity length), no pre-compression force is present. In this condition, the conical valve core responds instantaneously to any fluid-induced force. Conversely, if the spring length exceeds the cavity length, it becomes pre-compressed, introducing a preload. Under such conditions, the valve core remains stationary until the fluid pressure surpasses the pre-compression force. When the fluid force is below this threshold, the valve core does not move, and the system behaves analogously to a conventional viscous damper, maintaining a relatively constant damping coefficient.

2.1.2. Variable stiffness device

The variable stiffness device (Fig. 5) consists of a shell, a rod, disc springs (DS), and a base. Multiple sets of disc springs, with varying free heights and stiffness coefficients, are concentrically arranged along the rod to create a multi-stage stiffness mechanism.

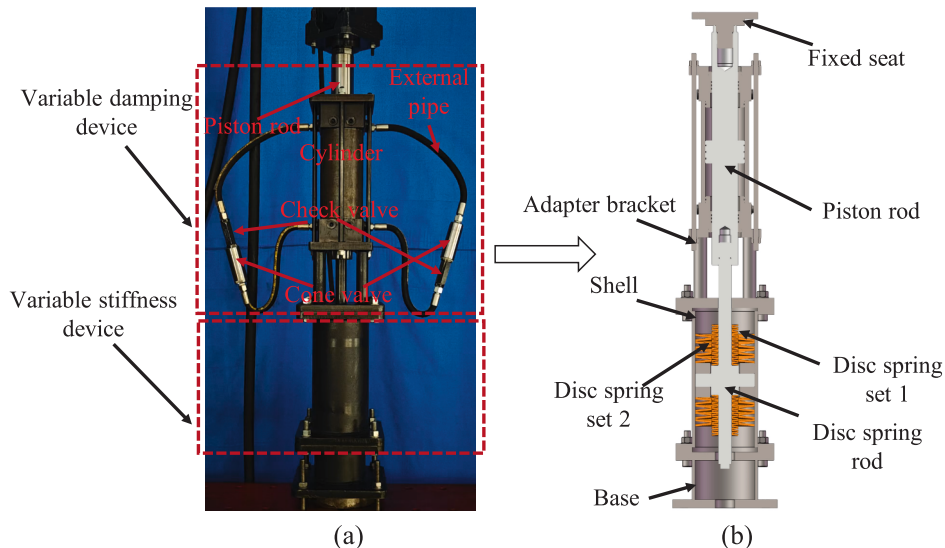


Fig. 2. Illustration of PVDSD: (a) PVDSD photo; (b) cutaway view.

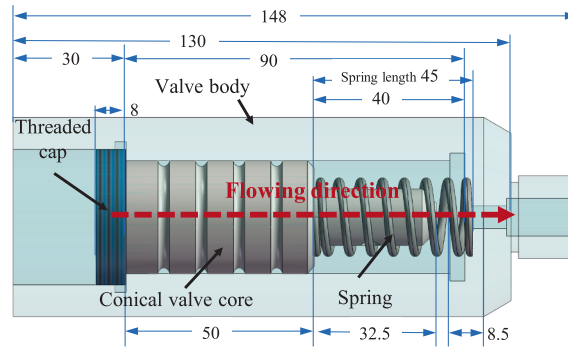


Fig. 3. Schematic diagram and dimensions of the cone valve (unit: mm).

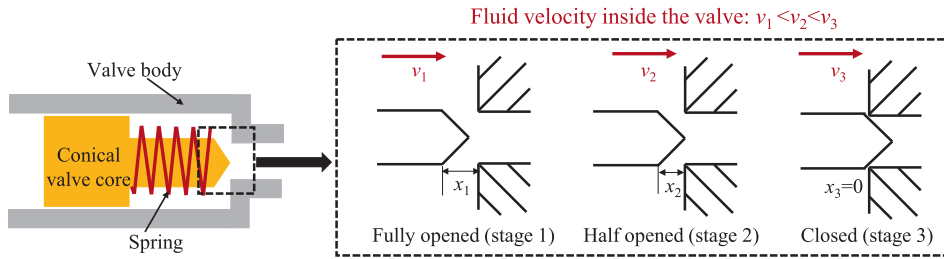


Fig. 4. Cone valve states under different flowing velocities.

These disc spring sets are symmetrically arranged on both sides of the rod. In this study, two sets of disc springs are employed to realize a two-stage variable stiffness. When the displacement remains within $\pm d_1$, the device remains inactive; the disc spring are in a preload-free state, as illustrated between Fig. 6(a) and Fig. 6(b), corresponding to negligible stiffness. Once the displacement exceeds the first-stage threshold ($\pm d_1$), the first set of disc springs begins to compress, producing a first-stage limiting stiffness force, as shown between Fig. 6(b) and Fig. 6(c). The stiffness in this range is governed by a single set of disc springs with a stiffness coefficient of k_1 . As the displacement further exceeds the second-stage threshold ($\pm d_2$), both sets of disc springs are compressed simultaneously. At this point, the stiffness is provided by the inner and outer layers of disc springs, resulting in a combined stiffness ($k = k_1 + k_2$) and generating the second-stage limiting stiffness force (Fig. 6(d)). Overall, the stiffness of the device increases in a stepwise manner with displacement, allowing it to adapt to varying loading demands efficiently.

2.2. Performance test of PVDSD

2.2.1. Tests program

Performance tests of PVDSD were conducted in the Structural Laboratory at Qingdao University of Technology. The test setup is

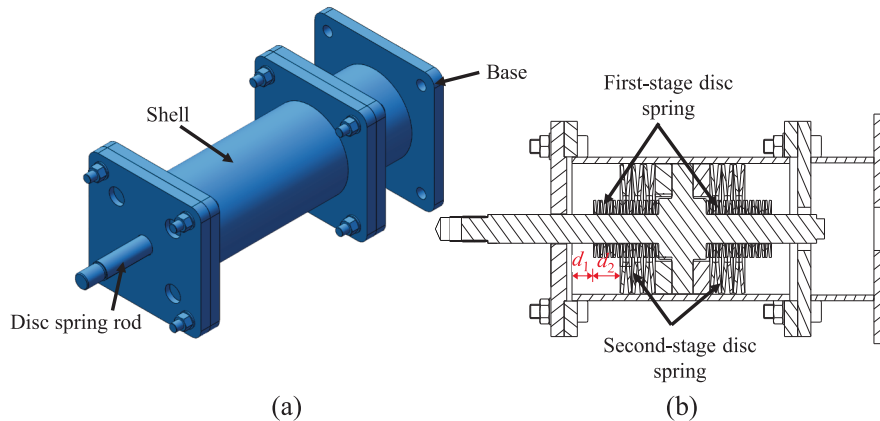


Fig. 5. Variable stiffness device: (a) device appearance; (b) internal diagram.

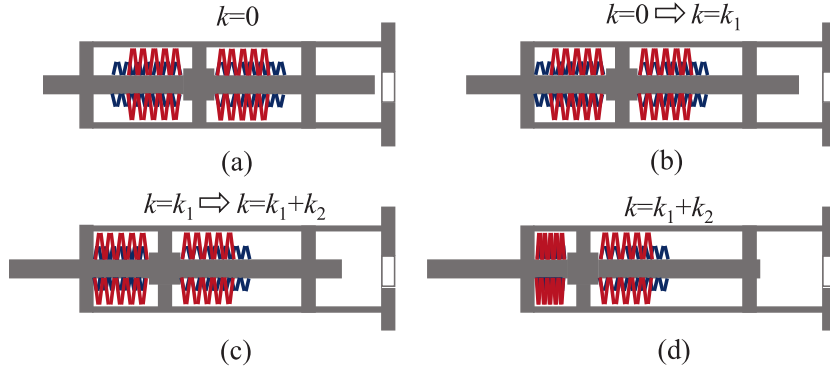


Fig. 6. Working states of the disc spring under different displacements: (a) initial uncompressed state; (b) first-stage compression; (c) second-stage compression; (d) fully compressed state.

illustrated in Fig. 7. A 10-ton actuator was employed to apply sinusoidal displacement inputs. Tests were performed under loading amplitudes of 30 mm, 50 mm, 70 mm, and 90 mm, with excitation frequencies gradually increased in increments of 0.02 Hz, as summarized in Table. 1. Each test condition was repeated for five cycles, with the third cycle selected for analysis [35]. Two disc spring

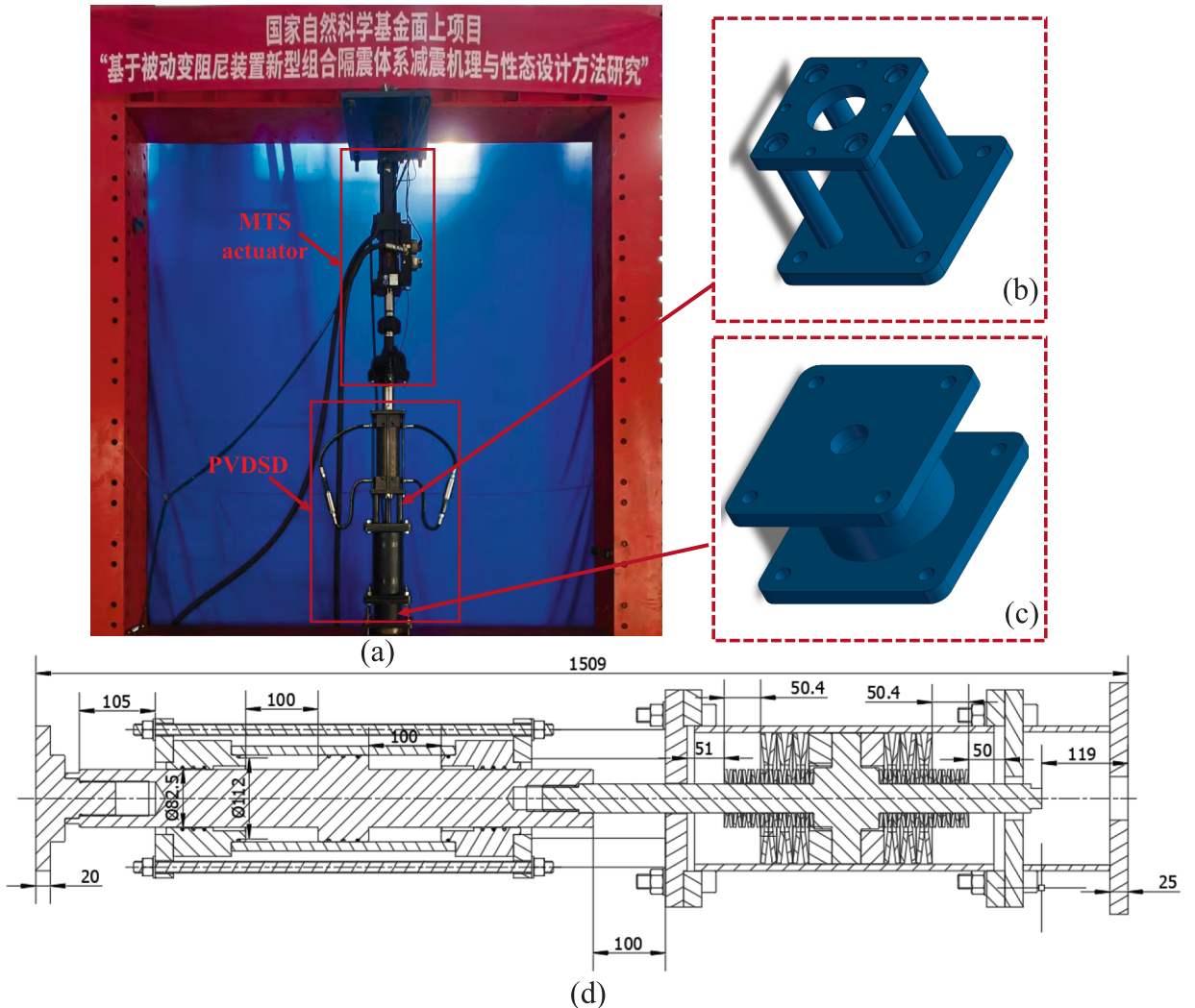


Fig. 7. Test device diagram: (a) test setup; (b) connection frame; (c) base; (d) PVDS dimensions (unit: mm).

combinations (DSCs) and two cone valve spring combinations are selected in the test. The spring parameters of the disc springs and cone valve springs are listed in Table 2 and Table 3, respectively. Detailed test cases and data specifications are also provided in Table 3. Data acquisition was recorded using embedded displacement sensors and force sensors with a sampling frequency of 50 Hz, while velocity was computed as the first derivative of the displacement signal.

2.2.2. Test result

(a) Overall behavior of PVDSD

PVDSD is characterized by variable damping and stiffness properties, making it essential to comprehensively evaluate its overall behavior. Fig. 8 and Fig. 9 depict the hysteresis curves of the variable damping and stiffness devices under configurations CB1 and CB2, respectively. It should be noted that due to the capacity of the 10-ton actuator, the force is capped at around 100 kN for safety purposes. Table 4 summarizes the decomposition of peak force corresponding to various displacement at 0.24 Hz.

As illustrated in Fig. 8 and Fig. 9, hysteresis curves exhibit pronounced nonlinear behavior, characterized by the plump loops and significant stiffness changes. Under small-amplitude loading of 30 mm and 50 mm, the variable stiffness property is inactivated due to the small loading displacement. While as the loading displacement increases to 70 mm and 90 mm, the force exhibits variable stiffness both in stage one and two respectively, especially at low-frequency loading where the velocity-related damping forces are small (Fig. 9 (a)). However, under high-frequency loadings, the response is dominated by damping effects, rendering the contribution of variable stiffness less significant (Fig. 9(f)).

Table 4 presents the values of stiffness and damping force corresponding to different displacement at a frequency of 0.24 Hz. The peak spring force is derived from the theoretical stiffness of the disc spring, while the peak damping force is obtained by subtracting the peak spring force from the total force. At displacement of 30 mm and 50 mm, only the damping force are observed, with no contribution from the spring component. As the displacement increases to 70 mm and 90 mm, the first and second-stage limiting devices are progressively activated, and the proportion of stiffness force increases to 25 % and 88 %, respectively. This trend indicates that at larger displacements, the stiffness device becomes the dominant contributor, while damping plays a supplementary role in energy dissipation. Conversely, at smaller displacements, damping dominates the response, serving as the primary means of energy dissipation.

Theoretically, the damping force reaches its peak when the displacement is zero. However, experimental results reveal a time lag caused by the presence of air bubbles within the damping cylinder. As the moving cone valve initially compresses the trapped gas, the peak damping force is shifted away from the zero-displacement position. This lag becomes more pronounced at higher velocities. Applying pre-pressure to the damping cylinder can alleviate this effect. Moreover, discrepancies in the maximum forces on either side of the damping valve are attributed to machining inaccuracies. Variations in friction between the moving valve core and its housing result in asymmetric displacement responses under identical loading conditions, leading to asymmetric damping forces. Such asymmetry can be reduced through enhanced machining precision and optimized surface treatments. The plateau regions observed in the experimental curves, where the damping force reaches the actuator's limit of 100 kN, predominantly occur under high loading velocities and large displacement amplitudes, indicating that the actuator has reached its maximum output capacity. Nevertheless, this does not affect the trends of variable damping and stiffness, nor the stability analysis of the PVDSD, and the conclusions remain robust. To comprehensively evaluate the damper's performance, future tests should utilize actuators with greater force capacity.

(b) Variable Stiffness Characteristics

At low loading velocities, the velocity-related damping force is negligible compared to the variable stiffness contribution. The deformation of the disc springs in the test remains within 75 % of their maximum deflection, a linear stiffness approximation is adopted to characterize their behavior. Fig. 10 presents the test results of PVDSD incorporating two disc spring combinations under their respective lowest loading frequencies. In Fig. 10, solid lines represent the theoretical predictions, while dashed lines correspond to the measured test results.

From Fig. 10, the stiffness restoring force exhibits a distinct stepwise increase with displacement. For instance, with the limitation

Table 1
Loading protocol.

Case	Frequency/Hz	Magnitude/mm	Maximum speed/(mm/s)
1	0.04	±30	7.54
2	0.04	±50	12.56
3	0.04	±70	17.58
4	0.04	±90	22.61
5	0.06	±30	11.30
6	0.06	±50	18.84
7	0.06	±70	26.38
8	0.06	±90	33.91
...	...	...	...
60	0.32	±90	180.86

Table 2
Disc spring parameters.

Combination	Model	Outer diameter D/mm	Inner diameter d/mm	Thickness t/mm	Deformation h_0/mm	Free height H_0/mm	Quantity	Combination configuration	Limit distance /mm
DSC1	Inner (DS1)	80	41	4	2.2	6.2	20	In series	$d_1 = 50.0$
	Outer (DS3)	180	92	6	5.1	11.1	8	In series	$d_2 = 28.2$
DSC2	Inner (DS2)	80	41	3	2.3	5.3	24	In series	$d_1 = 47.8$
	Outer (DS3)	180	92	6	5.1	11.1	8	In series	$d_2 = 31.4$

Table 3
Spring parameters of cone valve and Combinations of tested specimens.

Combination	Disc spring	Cone valve spring Spring of cone valve	Spring size (wire diameter $\times$ outer diameter $\times$ length) /(mm $\times$ mm $\times$ mm)	Material elastic modulus /GPa	Spring stiffness /(N/mm)	Pre-pressure force /N	Loading frequency /Hz	Loading displacement /mm
CB1	DSC1	S1	$4.0 \times 25 \times 45$	190	1.34	0	0.08 to 0.32	$\pm 30, \pm 50, \pm 70, \pm 90$
CB2	DSC2	S2	$4.5 \times 26 \times 45$	190	1.51	0	0.04 to 0.32	$\pm 30, \pm 50, \pm 70, \pm 90$

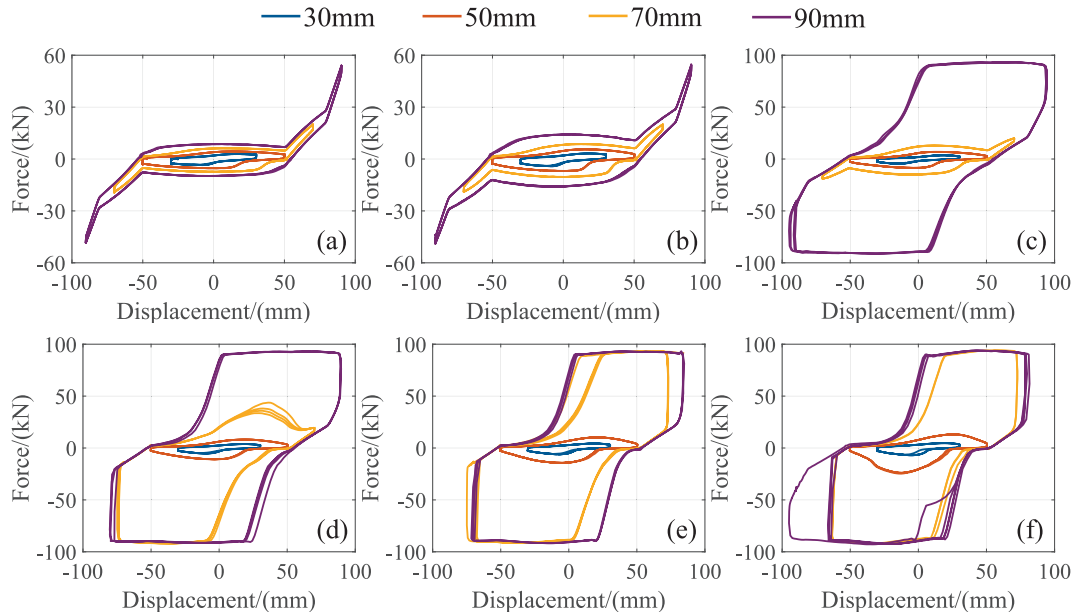


Fig. 8. Hysteretic curves of PVDSD with CB1: (a) 0.12 Hz; (b) 0.16 Hz; (c) 0.20 Hz; (d) 0.24 Hz; (e) 0.28 Hz; (f) 0.32 Hz.

distances as about 50.0 mm and 78.2 mm in DSC1, the zero, first stage, and second-stage stiffnesses are observed when loading amplitude as 30 and 50 mm, 70 mm, and 90 mm, respectively. The close agreements between test and theoretical results confirm the accuracy and reliability of the device's stiffness variation characteristics. However, due to fabrication tolerances and discrepancies between the actual and theoretical disc spring thicknesses, the maximum displacement observed in the experiments was slightly lower than the intended value, leading to a slightly lower measured stiffness at peak displacement. Overall, the experimental results validate that a well-designed variable stiffness system can effectively modulate stiffness within different displacement regimes.

(c) Variable damping characteristics

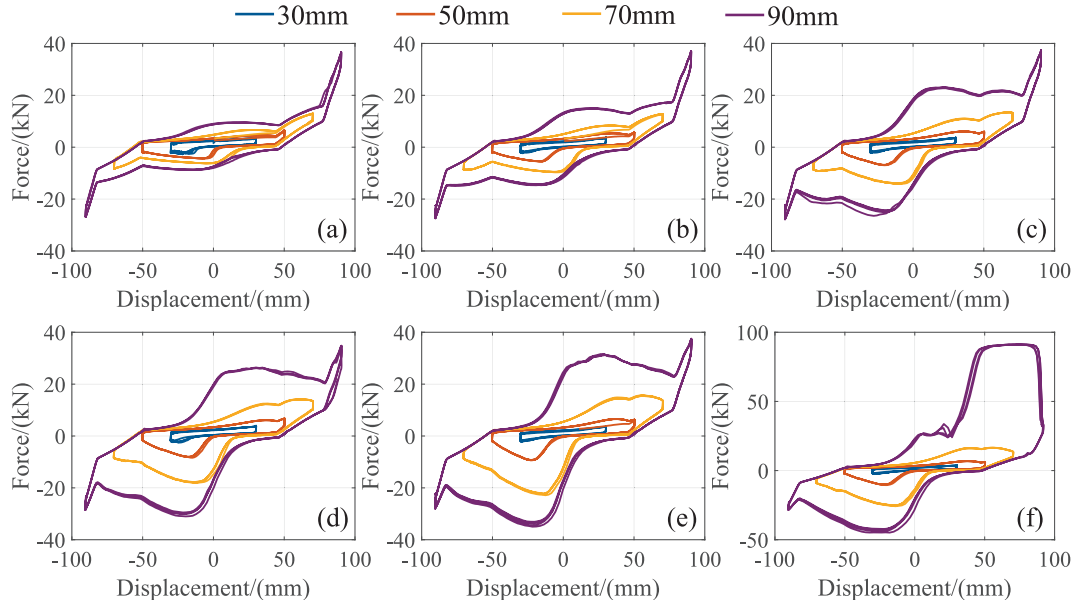


Fig. 9. Hysteretic curves of PVDSD with CB2: (a) 0.12 Hz; (b) 0.16 Hz; (c) 0.20 Hz; (d) 0.22 Hz; (e) 0.24 Hz; (f) 0.26 Hz.

Table 4

Peak force decomposition results at different displacement (0.24 Hz-90 mm).

Displacement/mm	Frequency/Hz	Force/kN	Peak spring force/kN	Peak damping force/kN	Spring force ratio/%	Damping force ratio/%
30	0.24	31.2	0	3.8	0	100
50	0.24	27.5	0	9.4	0	100
70	0.24	24.7	6.2	18.5	25	75
90	0.24	37.2	32.7	4.5	88	12

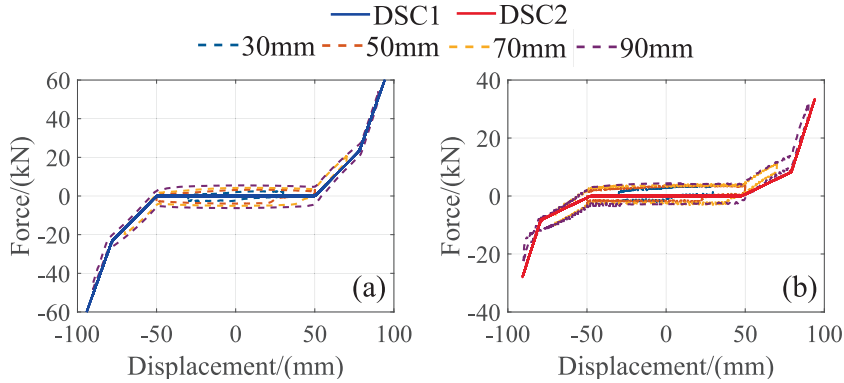


Fig. 10. PVDSD hysteretic curves at lowest velocities with (a) DSC1 at 0.04 Hz and (b) DSC2 at 0.08 Hz.

To investigate the variable damping characters, damping forces was isolated by subtracting the variable stiffness component from the total measured force. The resulting hysteresis curves of the variable damping force, under a loading displacement of ± 90 mm, are shown in Fig. 11.

As illustrated in Fig. 11, the hysteresis curves become more pronounced with rising loading frequency, indicating a significant enhancement in the system's energy dissipation capacity. Distinct differences in peak damping forces between Figs. 11(a) and (b), achieved by employing different spring configurations in the cone valves, confirm that the damping force is tunable through adjustments in spring stiffness. Under low frequencies (e.g., 0.08 Hz in CB1 and 0.12 Hz in CB2), the hysteresis curves exhibit relatively symmetrical profiles, attributed to the limited displacement range of the cone valve core and correspondingly low damping coefficients. However, as the frequency increases, significant time-lags emerge in the hysteresis loops, leading to the asymmetric force behavior between positive and negative displacements directions. Post-test inspections reveal that this asymmetry arises from

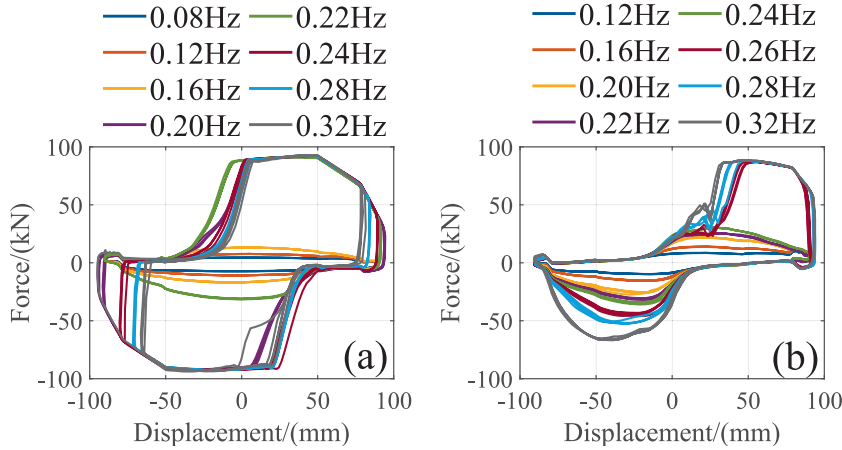


Fig. 11. Isolated variable damping force hysteretic curves: (a) CB1; (b) CB2.

increased friction between the cone valve body and core, as well as entrapped air bubbles in the damping fluid, both of which significantly delay the initial motion of the cone valve core.

Subsequently, to illustrate the characteristics of the variable damping force and the corresponding damping coefficient, the maximum damping forces are compared with the ideal linear viscous damper (LVD) at various velocities, as shown in Fig. 12. The LVD force is obtained by extrapolating the first two data point.

The divergence between variable and fixed damping forces underscores the pronounced nonlinearity of variable damping force and coefficient. This nonlinear damping behavior primarily arises from variation in the conical orifice area, induced by the motion of the valve core. As the velocity increases, the displacement of the valve core becomes larger, thereby reducing the conical opening and significantly enhancing the damping force. This mechanism enables the variable damping device to more effectively accommodate energy dissipation demands under high-speed conditions, delivering enhanced damping performance. Consequently, the variable damping system outperforms traditional linear damping systems in high-frequency or high-speed environments, better satisfying energy dissipation requirements at higher speeds.

3. Numerical models of LST and isolation system with PVDSD

3.1. Introduction of PVDSD isolation system

In conventional seismic isolation designs for LST, LRBs are commonly employed to mitigate seismic impacts. However, LRBs may experience excessive displacements beyond their design limits under high-intensity seismic excitations. Additionally, traditional isolation bearings exhibit low damping and limited energy dissipation capacity. To address these challenges, a novel seismic isolation system is proposed, as illustrated in Fig. 13. This system integrates a PVDSD into the conventional LRB-based isolation. The incorporation of the PVDSD enhances damping within the isolation layer, improves energy dissipation, and provides adaptive stiffness constraints under extreme seismic conditions. Under low-intensity earthquakes, the isolation layer experiences small displacements, and energy dissipation is primarily provided by the LRB. As the seismic intensity increases to moderate intensities, the variable damping capability of PVDSD supplies additional damping to reduce structural responses. While, during high-intensity earthquakes, the variable stiffness device delivers substantial restoring stiffness, effectively constraining excessive displacements and safeguarding

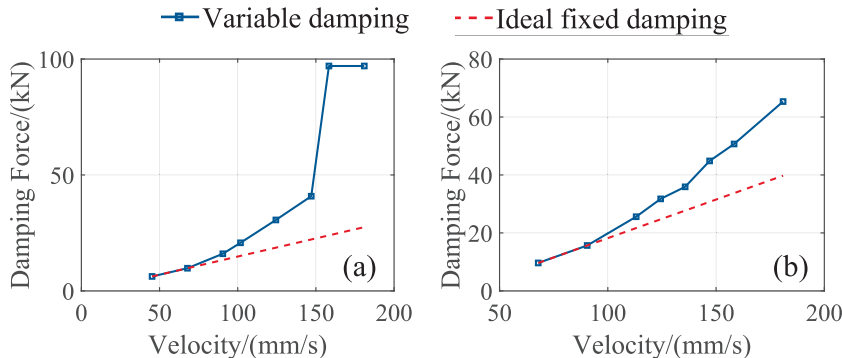


Fig. 12. PVDSD damping force-velocity relationship compared with ideal LVD: (a) PVDSD with CB1; (b) PVDSD with CB2.

the structure against collapse.

The proposed isolation system consists of a LRB and a PVDSD in parallel, ensuring that both components are subjected to identical displacements and velocities. The LRB is positioned at the base of the LST, directly anchored to the ground, effectively isolating seismic waves and mitigating horizontal displacements to preserve structural stability. Meanwhile, the PVDSD adaptively modulates its damping and stiffness characteristics to further enhance energy dissipation.

3.2. Numerical model and dynamic equation of LST

3.2.1. Numerical model of LRB

The isolation bearing employed in this study is the LRB, whose nonlinear behavior is modeled using the Bouc-Wen model, which closely approximates the actual hysteresis curve [36]. The mathematical expression for the restoring force (F_b) of LRB is mathematically expressed as:

$$F_b = \alpha \frac{F_y}{D_y} x + (1 - \alpha) F_y Z \quad (1)$$

where x is the displacement applied to the LRB; α represents the stiffness ratio of the bearing, defined as the ratio of the post-yield stiffness to the elastic stiffness; F_y and D_y denote the yield force and yield displacement of the bearing, respectively; The hysteretic component Z characterizes the bearing's energy dissipation behavior, which is defined by the following first-order differential equation:

$$D_y \dot{Z} = A \dot{x} - \gamma |\dot{x}| |Z|^{\eta-1} - \beta \dot{x} |Z|^\eta \quad (2)$$

where, A controls the hysteresis loop area; $\dot{x}$ is the velocity of the LRB; β and γ control the shape of the hysteretic loop; η controls the smoothness of the hysteretic curve. The relevant parameters A , γ , β and η of the Bouc-Wen model, which characterizes the nonlinear behavior of the isolation layer, are taken from Reference [37], and assigned values of 1, 0.5, 0.5, and 1.0, respectively. To verify the accuracy of the developed Bouc-Wen model, the hysteretic curve of the numerical model was compared with the experimental results reported in [38], as shown in Fig. 14. The good agreements between the developed Bouc-Wen model and test results demonstrates the reliability of selected model in representing the hysteretic behavior of the LRB.

3.2.2. Mechanical model of PVDSD

The mechanical model of the PVDSD is illustrated in Fig. 15, with the control force expressed as follows:

$$F_d = F_v + F_s \quad (3)$$

where, F_d represents the total force generated by the PVDSD, F_v denotes the damping force, and F_s corresponds to the stiffness force, x and v are the displacement and velocity of the device, respectively.

(a) Numerical model of variable stiffness device

According to the China standard for disc springs [39], the load-deformation relationship for a single disc spring without a

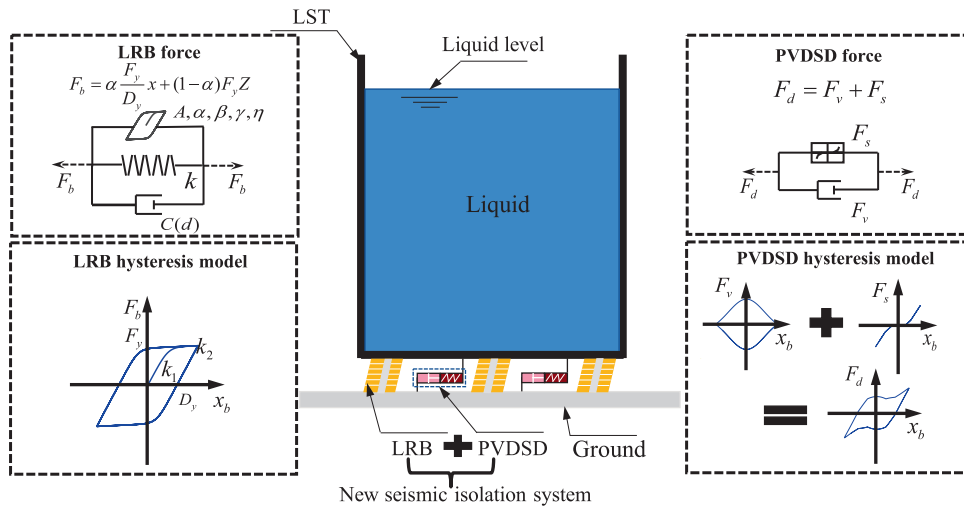


Fig. 13. PVDSD-LRB isolation system for LST.

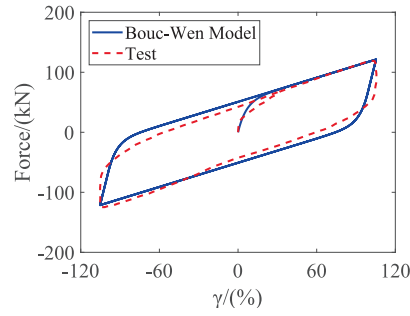


Fig. 14. Comparison between the model and experimental results.

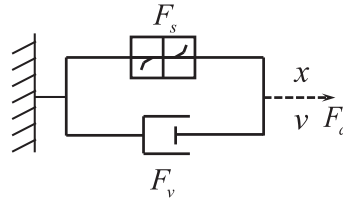


Fig. 15. Mechanical model of PVDSD.

supporting surface is expressed as:

$$f_s = \frac{4E}{1-\mu^2} \cdot \frac{t^4}{K_1 D^2} \cdot \frac{\delta}{t} \cdot \left[\left(\frac{h_0}{t} - \frac{\delta}{t} \right) \left(\frac{h_0}{t} - \frac{\delta}{2t} \right) + 1 \right] \quad (4)$$

where, f_s represents the reacting force; E is the elastic modulus; μ denotes Poisson's ratio; t corresponds to the thickness of the disc spring; D and d are the outer and inner diameter of the disc spring, respectively; h_0 is the maximum deflection when the disc spring is fully flattened; δ denotes the deformation of a single disc spring. Additionally, K_1 is an intermediate variable, expressed as follows:

$$K_1 = \frac{1}{\pi} \cdot \frac{\left[\frac{C-1}{C} \right]}{\frac{C+1}{C-1} - \frac{2}{\ln C}} \quad (5)$$

$$C = \frac{D}{d} \quad (6)$$

When the assembly consists of multiple disc springs arranged in series, the resultant force of the disc spring assembly is expressed as:

$$F_s = f_s \quad (7)$$

$$\delta_1 = n \cdot \delta \quad (8)$$

$$k = F_s / \delta_1 \quad (9)$$

where, F_s represents the load of the disc spring assembly; n represents the number of disc springs in series; δ_1 denotes the total deformation of the disc spring assembly; k is the equivalent linear stiffness of the disc spring assembly.

When the deformation of the disc spring assembly remains within 75 % of its maximum capacity, its stiffness can be approximated as linear. Accordingly, the stiffness force generated by the n -stage variable stiffness device is expressed as:

$$F_s = \begin{cases} 0 & x \leq d_1 \\ k_1(x - d_1) & d_1 < x < d_2 \\ k_1 d_2 + (k_1 + k_2)(x - d_1 - d_2) & d_2 < x < d_3 \\ \vdots & \\ (k_1 + k_2 + \dots + k_{n-1})(d_2 + d_3 + \dots + d_n) + (k_1 + k_2 + \dots + k_n)(x - d_1 - d_2 - \dots - d_n) & d_n < x < d_{n+1} \end{cases} \quad (10)$$

In the current study, a two-stage variable stiffness device is constructed using two sets of disc springs. Its numerical model can be expressed as:

$$F_s = \begin{cases} 0 & x \leq d_1 \\ k_1(x - d_1) & d_1 < x < d_2 \\ k_1 d_2 + (k_1 + k_2)(x - d_1 - d_2) & x \geq (d_1 + d_2) \end{cases} \quad (11)$$

where, d_1 and d_2 represent the limiting distances of the first and second stages, respectively, while k_1 and k_2 denote the corresponding stiffness coefficients.

The theoretical stiffness-displacement relationships of the DSC1 and DSC2 disc spring assemblies are presented in Fig. 16 and summarized in Table. 5.

(b) Numerical model of variable damping device

The damping force generated by the cone valve is used to represent the overall damping force of the device, with the contributions from other components considered negligible. An analytical expression for the damping force as a function of velocity is established. Fig. 17 illustrates the schematic of the cone valve.

The flow area of the cone valve is denoted by A , and the displacement of the cone valve core induces a pressure difference ΔP between the inlet and outlet. The damping force generated by the cone valve due to orifice contraction is expressed as:

$$F = \Delta P \cdot A \quad (12)$$

$$A = \pi d x \sin \alpha \left(1 - \frac{x}{2d} \sin 2\alpha\right) \quad (13)$$

When the valve opening is neglected, the flow area A of the cone valve is given by [40]:

$$A = \pi d x \sin \alpha \quad (14)$$

$$d = \frac{d_1 + d_2}{2}$$

The flow equation of the cone valve orifice is:

$$Q = A \cdot v_c = C_d \pi d x \sin \alpha \sqrt{2 \Delta P / \rho} \quad (15)$$

where, C_d is the flow coefficient, and ρ is the fluid density; v_c is the velocity at conical damping orifice, which can be related to the piston rod velocity v via the principle of flow equivalence:

$$v_c = \frac{(A_{dc} - A_{rod})}{A} \cdot v = \mu \cdot v \quad (16)$$

$$\mu = \frac{(A_{dc} - A_{rod})}{A} \quad (17)$$

where, A_{dc} represents the cross-sectional area of the cylinder; A_{rod} denotes the cross-sectional area of the piston rod; Accordingly, the damping force generated by the conical valve can be expressed as:

$$F_v = \frac{\mu^2 v^2 \rho}{2 C_d^2} \pi d x(v) \sin \alpha \quad (18)$$

where, x is a function of velocity, k is a constant.

$$x(v) = k \cdot v^n \quad (19)$$

At the initial displacement x_0 , the damping coefficient is c_0 . When the displacement x varies with velocity, the damping coefficient evolves to c_v :

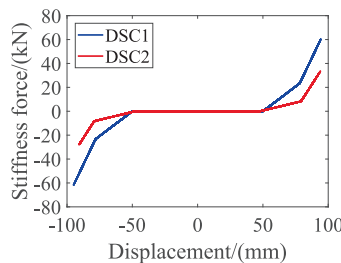
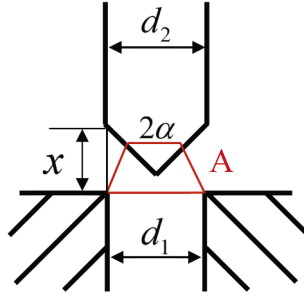


Fig. 16. Stiffness-displacement characteristics of disc springs.

Table 5

Disc spring parameters.

Model	Disc spring stiffness /(kN/mm)		Limit distance /mm	
	k_1	$k_1 + k_2$	d_1	$d_1 + d_2$
DSC1	0.796	2.21	50	78.2
DSC2	0.264	1.70	47.8	79.2

**Fig. 17.** Cone valve schematic diagram.

$$c_0 = \frac{\mu^2 \rho}{2C_d^2} A(x_0) \quad (20)$$

$$c_v(v) = \frac{\mu^2 \rho}{2C_d^2} A(x(v)) \quad (21)$$

Substituting Equation (14) and (19) into Equation (21):

$$c_v(v) = \frac{\mu^2 \rho}{2C_d^2} \cdot \pi d \cdot k \cdot v^n \sin \alpha = c_1 \cdot v^n \quad (22)$$

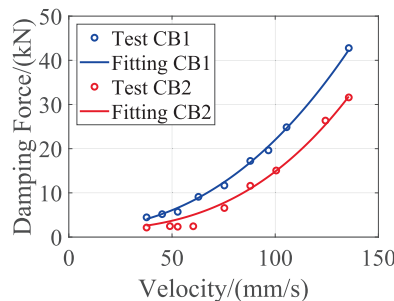
$$c_1 = \frac{\mu^2 \rho}{2C_d^2} \cdot \pi d k \sin \alpha$$

The relationship between the damping force F_v and velocity v is derived as:

$$F_v = \left\{ [c_1 \cdot (|v|)^n + c_0] \cdot (|v|)^\beta + F_f \right\} \cdot \text{sgn}(v) \quad (23)$$

where, F_f represents the friction force of the device. Both β and n are exponent parameters used to adjust the nonlinear characteristics of the viscous damping.

The experimental data were fitted to the theoretical model, and the resulting fitting curve depicted in Fig. 18. The corresponding fitting parameters are summarized in Table. 6. The strong agreement between test results and the numerical model validates the applicability of the model in capturing the nonlinear relationship between the damping force and input velocity.

**Fig. 18.** Damping coefficient fitting.

3.3. Numerical models of LST

The numerical model of the proposed seismic isolation system for LST is developed based on Housner-Haroun's elastic theory [41], as shown in Fig. 19. The liquid mass was simplified into convective mass (m_c), liquid–solid coupling mass (m_i) and rigid mass (m_0) in the storage tank. The convective mass and liquid–solid coupling mass are connected to the tank wall by equivalent spring stiffness k_c , k_i and damping coefficients c_c , c_i respectively. Additionally, the stiffness of the isolation bearing and the PVDSD are represented by k_0 and $K_d(x_0)$, respectively, with corresponding damping coefficients c_0 and $C_d(x_0)$. The equivalent heights of the convective, the liquid–solid coupling, and the rigid masses are denoted by h_c , h_i and h_0 , respectively. x_0 , x_i and x_c are the displacements of elements m_0 , m_i and m_c , respectively. $\dot{x}_0$, $\dot{x}_i$ and $\dot{x}_c$ are the velocity of masses, $\ddot{x}_c$, $\ddot{x}_i$ and $\ddot{x}_0$ represent the accelerations of the three masses, respectively, $\ddot{x}_g$ is the ground motion acceleration.

The dynamic equation of the system is derived using D'Alembert principle [42], where F_d denotes the combined effect of variable damping and variable stiffness forces acting on the isolation layer:

$$\begin{aligned} \begin{pmatrix} m_c & 0 & 0 \\ 0 & m_i & 0 \\ 0 & 0 & m_0 \end{pmatrix} \begin{pmatrix} \ddot{x}_c(t) \\ \ddot{x}_i(t) \\ \ddot{x}_0(t) \end{pmatrix} + \begin{pmatrix} c_c & & -c_c \\ & c_i & -c_i \\ -c_c & -c_i & c_c + c_0 + c_i \end{pmatrix} \begin{pmatrix} \dot{x}_c(t) \\ \dot{x}_i(t) \\ \dot{x}_0(t) \end{pmatrix} + \begin{pmatrix} k_c & & -k_c \\ & k_i & -k_i \\ -k_c & -k_i & \alpha k_0 + k_c + k_i \end{pmatrix} \begin{pmatrix} x_c(t) \\ x_i(t) \\ x_0(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ (1 - \alpha)k_0 \end{pmatrix} Z \\ = - \begin{pmatrix} m_c \\ m_i \\ m_0 \end{pmatrix} \ddot{x}_g(t) - \begin{pmatrix} 0 \\ 0 \\ F_d \end{pmatrix} \end{aligned} \quad (24)$$

The equations for base shear force Q , overturning moment M , and sloshing wave height h_v are respectively given as follows:

$$Q = m_0[\ddot{x}_0(t) + \ddot{x}_g(t)] + m_i[\ddot{x}_g(t) + \ddot{x}_i(t)] + m_c[\ddot{x}_g(t) + \ddot{x}_c(t)] \quad (25)$$

$$M = m_0 h_0[\ddot{x}_0(t) + \ddot{x}_g(t)] + m_i h_i[\ddot{x}_g(t) + \ddot{x}_i(t)] + m_c h_c[\ddot{x}_g(t) + \ddot{x}_c(t)] \quad (26)$$

$$h_v = 0.837 \frac{\ddot{x}_c + \ddot{x}_g}{g} \quad (27)$$

4. Seismic structural response comparison of LST with different isolation systems

4.1. Structural parameters of LST

A large-scale LST with a capacity of 15000 m³ is selected for analysis. The liquid has a density (ρ) of 1000 kg/m³, and the storage tank has an elastic modulus of $E_s = 206$ GPa. The tank wall height H , radius R and liquid height H_1 are 21.7 m, 50 m and 20.1 m respectively. The key parameters of the storage tank are summarized in Table. 7. The damping ratios for the isolation layer, the solid–liquid coupled mass, and the convective sloshing mass are set to 0.2, 0.02, and 0.005, respectively [43–45]. The natural circular frequencies of the convective and impulsive components, as well as their spring–dashpot connections to the tank wall, were determined according to Reference [46].

4.2. Ground motion selection

Seven ground motion records are selected from the NGA-West2 database provided by the Pacific Earthquake Engineering Research Center (PEER). The characteristics of the selected representative seismic waves are summarized in Table. 8, and their corresponding acceleration spectra with damping ratio of 5 % are illustrated in Fig. 20.

Taking the peak ground acceleration (PGA) as the intensity measure (IM), the ground motions are scaled to four hazard levels: Frequent Earthquake (FE), Design-Based Earthquake (DBE), and Maximum Considered Earthquake (MCE), Rare Earthquake (RE), with corresponding intensities are 0.14 g, 0.4 g, 0.62 g, and 1.0 g, respectively [47].

Table 6
Damping coefficient fitting parameters.

Case	β	Constant damping coefficient $c_0 / (\text{kN} \cdot (\text{s}/\text{mm})^\beta)$	Variable damping parameter $c_1 / (\text{kN} \cdot (\text{s}/\text{mm})^{\beta+n})$	Exponent n	Friction force F_f / kN
CB1	0.451	0.121	0.00056	1.821	1.213
CB2	0.450	0.103	0.00012	2.073	1.201

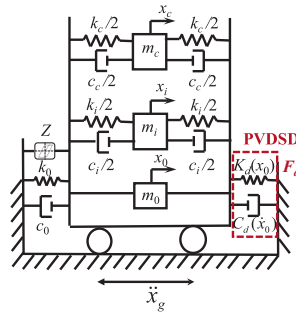


Fig. 19. Numerical model of new seismic isolation system tank.

Table 7

Structural parameters of LST.

H_1/R	m_0/m_l	m_l/m_l	m_c/m_l	h_0/H_1	h_l/H_1	h_c/H_1
0.402	0.08	0.2038	0.7147	0.902	0.4160	0.5242

4.3. Isolation system parameters

To evaluate the effectiveness of the proposed method in controlling liquid sloshing within a rigid storage tank, four isolation strategies are considered and implemented: (1) LRB, (2) LRB combined with a linear viscous damper (LRB + LVD), (3) LRB combined with a variable damping viscous damper (LRB + VDVD), and (4) LRB integrated with a passive variable damping and stiffness device (LRB + PVDS). For comparative purposes, the response of the uncontrolled (NC) LST is computed.

The parameters of the isolation layer and associated devices are summarized in Table. 9. The stiffness and damping of the isolation layer are obtained from 40 LRB400 units. The damping coefficient of the PVDS is contributed by 60 PVDS units, while the parameters c_0 , c_1 , stiffness, and limit distance are determined through iterative optimization calculations. Three configurations (LRB + LVD, LRB + VDVD, and LRB + PVDS), incorporate different damping devices into the LRB system. All three configurations share the same constant damping coefficient c_0 , while both the LRB + VDVD and LRB + PVDS systems additionally employ an identical variable damping coefficient c_1 .

4.4. Result and discussion

Five relevant engineering demand parameters including base shear, overturning moment, isolation layer displacement, sloshing wave height, and tank wall pressure are considered in current study. To assess the vibration control efficacy of various structural systems under different seismic hazard levels, a response reduction factor R_f is established:

$$R_f = \frac{Q_1 - Q_2}{Q_2}$$

where, R_f denotes the response reduction factor of control system Q_1 relative to control system Q_2 , with Q representing either the average peak value or the peak value.

4.4.1. Displacement of isolation layer

To investigate the displacement mitigation effectiveness of four isolation systems, the time-history responses of the isolation layer displacements under four seismic hazard levels are illustrated in Fig. 21, where the results of the Imperial Valley, CHICHI, and El Centro ground motions are presented. The remaining four ground motions exhibit similar trends. The hysteresis curves of different control systems subjected to the CHICHI ground motion are depicted in Fig. 22. Fig. 23 presents corresponding R_f with respect to LRB system at four hazard levels.

As shown in Fig. 21, the peak displacement of the isolation layer increases with the seismic hazard level. At the FE level, all four isolation systems (LRB, LRB + LVD, LRB + VDVD, LRB + PVDS) exhibit similar displacement responses, with differences within 1.5 cm, indicating comparable performance. Moreover, the identical hysteretic curves of the LRB + VDVD and LRB + PVDS systems shown in Fig. 22 (a), suggest equivalent energy dissipation capacities and confirm the inactivation of variable stiffness mechanism at this hazard level.

At the DBE level, differences among the isolation systems become more pronounced. The LRB system shows the largest displacement (up to 15.4 cm in Fig. 21 (c)), while the LRB + PVDS system shows the smallest (approximately 3.5 cm), due to its additional variable stiffness and strong energy dissipation. While, although LRB + VDVD system is equipped with variable damping, it still exhibits relatively large residual displacements due to the lacking in extra variable stiffness. As the hazard level increases to the MCE and RE levels, significant residual displacements are observed in all control systems except for LRB + PVDS, highlighting its

Table 8
Characteristics of the ground motions.

Record No.	Earthquake	Year	Station	Magnitude	PGV (cm/s)	PGA (g)	Epicentral Distance (km)
1	Imperial Valley	1979	USGS 412 El Centro Array #10	6.53	47.07	0.21	26.31
2	CHICHI	1999	CWB TCU075	7.62	56.69	0.30	20.67
3	El Centro	1940	IVID, El Centro Array #9	6.90	33.30	0.35	12.20
4	Northridge	1994	CDMG 24436 Tarzana – Cedar Hill A	6.69	96.00	1.66	5.41
5	MorganHill	1984	CDMG 57191 Halls Valley	6.19	23.20	0.13	3.94
6	Irpinia	1980	Sturmo	6.90	71.10	0.23	10.80
7	Landers	1992	Lucerne	7.30	132.30	0.71	2.20

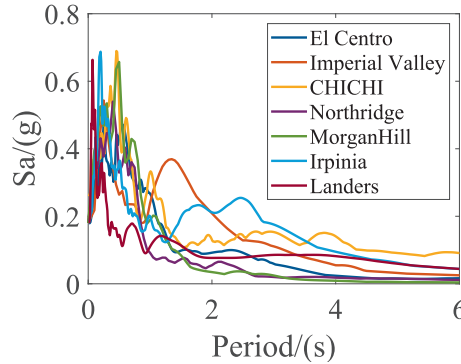


Fig. 20. Elastic acceleration response spectra of the ground motions.

Table 9
Parameters of seismic isolation system and devices.

Parameter	Value
Isolation Layer Stiffness/($\times 10^2$ kN/m)	Pre-yield: 232 Post-yield: 18
Isolation Layer Damping/($\times 10^2$ kN·s/m)	946
Damping Coefficient c_0 /(kN·(s/m) $^\beta$)	LRB + LVD: 33192 LRB + VDVD: 33192 LRB + PVDSD: 33192
Damping Coefficient c_1 /(kN·(s/m) $^{\beta+n}$)	LRB + VDVD: 200874 LRB + PVDSD: 200874
β	1
n	1.8
Stiffness/($\times 10^2$ kN/m)	First stage (k_1): 600 Second level ($k_1 + k_2$): 700
Limitation Distance/mm	First level (d_1): 12 Second level ($d_1 + d_2$): 17

superior self-centering and restoring capabilities.

Fig. 23 shows that under RE conditions, the LRB + PVDSD system achieves the highest displacement reduction (61.8 %), followed by LRB + VDVD (29.9 %) and LRB + LVD (13.7 %). The PVDSD device consistently shows superior and stable performance across all hazard levels, attributed to its variable damping and variable stiffness mechanisms.

4.4.2. Base shear and overturning base moment

Base shear and overturning moment are critical parameters for evaluating the structural stability of the LST. Fig. 24 presents the time-history responses of base shear and overturning moment under the Imperial Valley ground motion. Fig. 25 presents the corresponding peak values under the Imperial Valley, CHICHI, and El Centro ground motions. Table. 10 quantifies the average response reduction ratios of each isolation system relative to the NC system.

As shown in Fig. 24, all four isolation systems significantly reduce base shear and overturning moment, while effectively lengthening the structural vibration period compared to the NC system, confirming their efficacy in mitigating seismic responses of the LST. Fig. 25 further show that although both base shear and overturning moment increase with seismic hazard level, the isolation systems still provide notable reductions.

However, shear forces and overturning moments of hybrid systems (with LVD, VDVD, and PVDSD) are higher compared to the pure LRB isolation system. This observation aligns with the findings of Sun et al. [48], indicating that the enhanced displacement control capability of the hybrid systems is achieved at the expense of moderately increased base forces. Nevertheless, all isolation systems still significantly outperform the NC system. As shown in Table. 10 under the RE level, the base shear and overturning moment reduction ratios of the LRB + PVDSD system reach 80.1 % and 81.8 %, respectively. Therefore, although the LRB + PVDSD system may introduce slightly higher base responses, it achieves superior displacement mitigation, particularly under severe seismic events (MCE and RE).

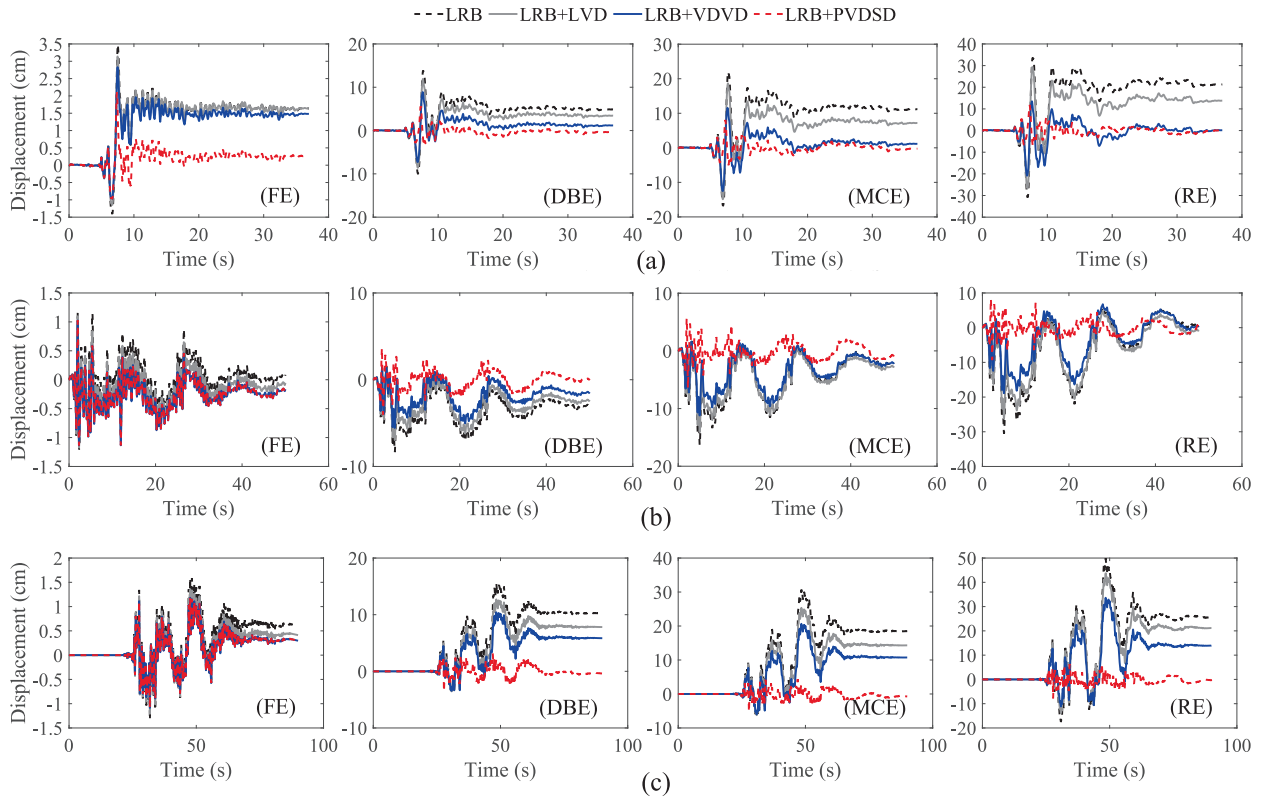


Fig. 21. Isolation layer displacement under different hazard levels: (a) Imperial Valley; (b) El Centro; (c) CHICHI.

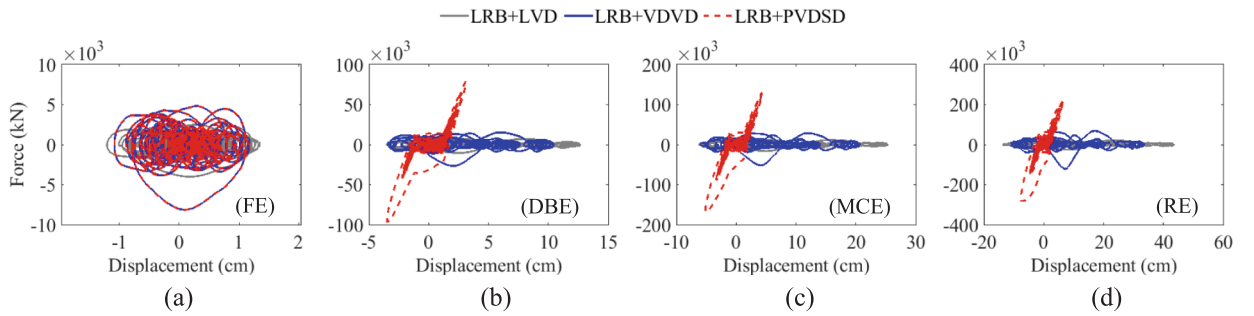


Fig. 22. Hysteresis curve (CHICHI).

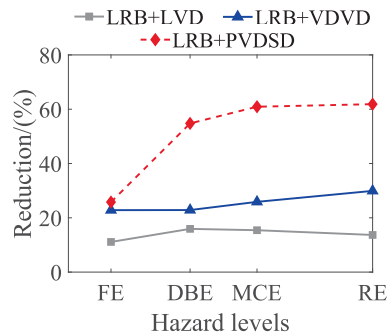


Fig. 23. Average reduction ratio of isolation displacement relative to that of the LRB system.

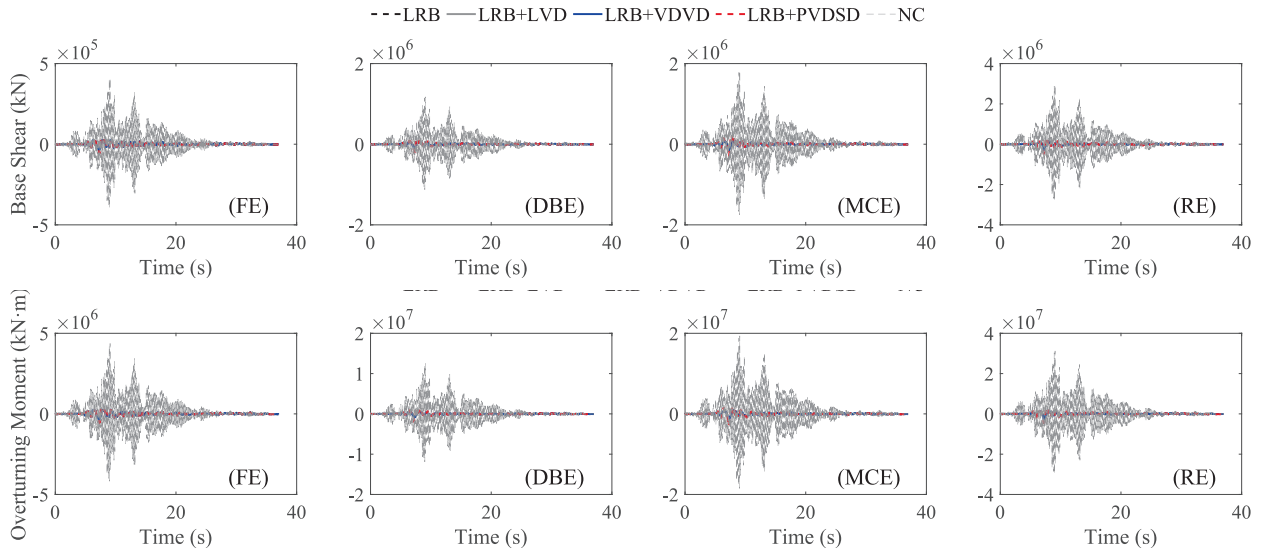


Fig. 24. Base shear and overturning moment of Imperial Valley at different hazard levels.

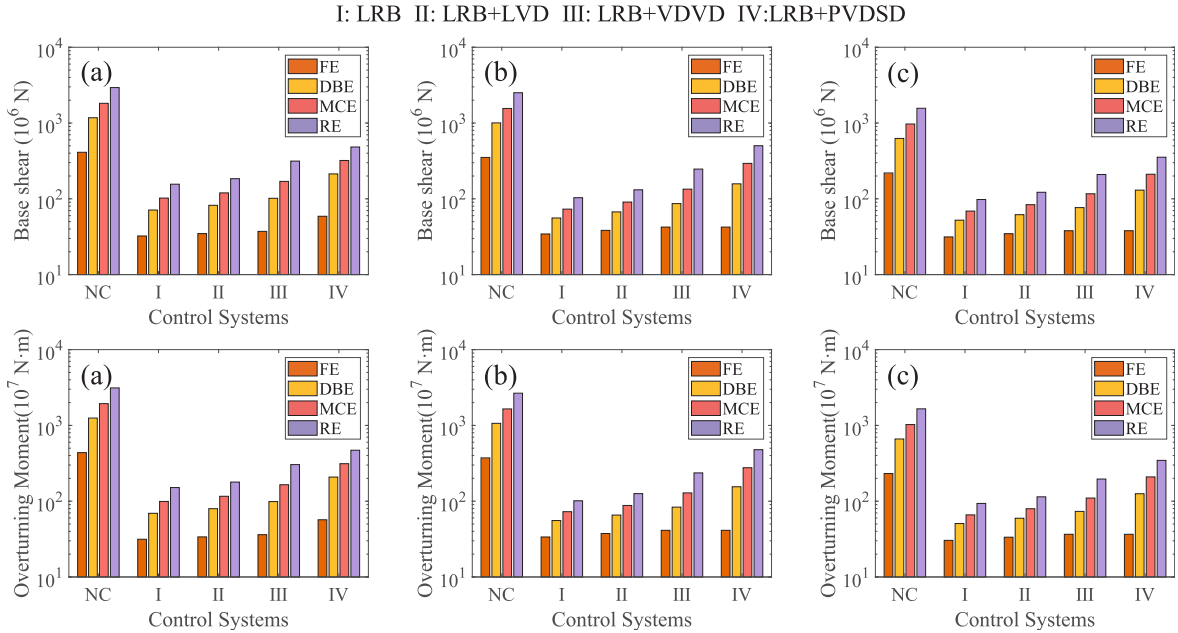


Fig. 25. Effects of control systems on base shear and overturning moment under seismic excitation: (a) Imperial Valley; (b) El Centro; (c) CHICHI.

Table 10

Average reduction rates of base shear and overturning moment achieved by LRB + PVDSD relative to the NC system.

Hazard levels	Base shear				Overturning moment			
	FE	DBE	MCE	RE	FE	DBE	MCE	RE
$R_f / \%$	84.7	80.4	79.8	80.1	86.1	82.0	81.6	81.8

Overall, the LRB + PVDSD isolation system offers a more optimal balance between displacement control and structural safety.

4.4.3. Sloshing height

In the seismic response analysis of base-isolated LST, the sloshing wave height is a critical indicator of internal fluid dynamics.

Fig. 26 shows time-history responses of sloshing wave height under four hazard levels for the Imperial Valley, CHICHI, and El Centro ground motions, while Table. 11 summarizes the average reduction ratios of sloshing wave height achieved by different systems relative to the NC system.

As shown in Fig. 26, the LRB, LRB + LVD, and LRB + VDVD systems exhibit similar sloshing wave heights across all hazard levels. In comparison with other damping-enhanced isolation systems, the LRB + PVDSD configuration exhibits a slightly greater sloshing response. As shown in Table. 11 under FE seismic intensity, all isolation systems generate a counterproductive effect on sloshing wave height due to the relatively small amount of damping provided. Same trends were also observed by Malhotra [7]. Under DBE, MCE, and RE seismic intensities, however, the LRB + PVDSD system demonstrates a negative effect relative to the other isolation systems. This phenomenon arises because LRB, LRB + VDVD, and LRB + VDVD systems all possess substantial energy dissipation capacity, while the additional stiffness introduced by the PVDSD shortens the natural period of the isolated LST. This, in turn, amplifies the acceleration response of the LST and consequently increases the sloshing wave height. Notably, at the RE level, as illustrated in Fig. 26 (b), the peak sloshing wave height for the LRB + PVDSD system reaches 28.8 cm, representing a 57.4 % increase compared with the LRB system (18.3 cm). Nevertheless, this increase is limited and does not endanger the safety of the storage tank. For instance, under FE ground motions, the sloshing wave height increases by only about 1 %. In scenarios such as MCE, where large deformation of the isolation layer occurs, the primary objective is to control displacement to prevent overturning, while the slight increase in sloshing can be further mitigated through the use of auxiliary anti-sloshing devices [49,50].

4.4.4. Peak hydrodynamic pressure

Under seismic excitation, the external forces exerted on the walls of a LST primarily originate from hydrodynamic pressures. Based on the hydrodynamic pressure calculation method summarized in reference [51], Fig. 27 illustrates the distribution of hydrodynamic pressure along the tank wall height under Imperial Valley ground motion for different control systems at four hazard levels. Fig. 28 further summarizes the average peak hydrodynamic pressure on the tank wall under selected earthquake records.

As shown in Fig. 27, the maximum hydrodynamic pressure occurs at the tank bottom, primarily attributed to the combined effects of fluid inertia and structural confinement. Compared with the NC system, all four isolation systems significantly reduce this pressure, with an average reduction of 46.7 % at the RE level.

As evidenced in both Fig. 27 and Fig. 28, the differences in hydrodynamic pressure reduction among the various control systems are marginal. The additional variable stiffness introduced by the PVDSD results in a slight increase in response. All the evidence indicates that the isolation system plays an essential role in decreasing hydrodynamic pressures, while the influence of supplemental damping remains limited.

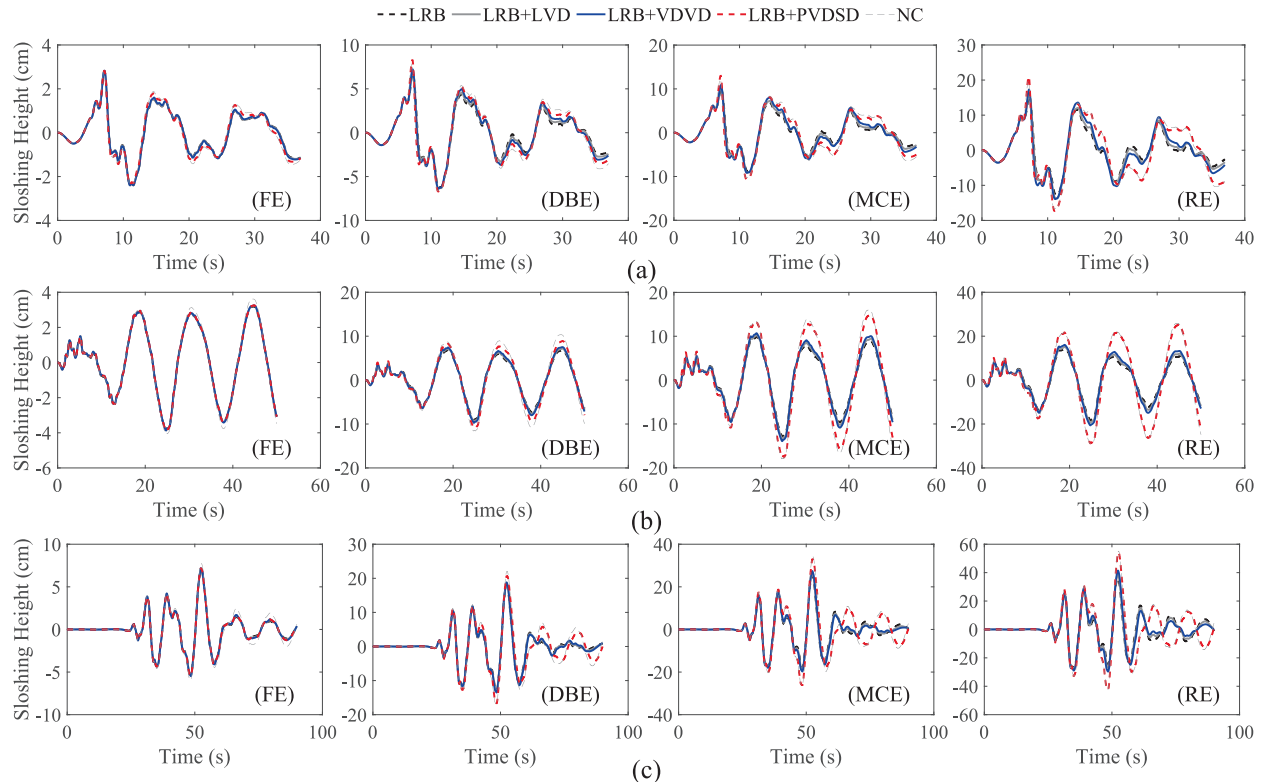


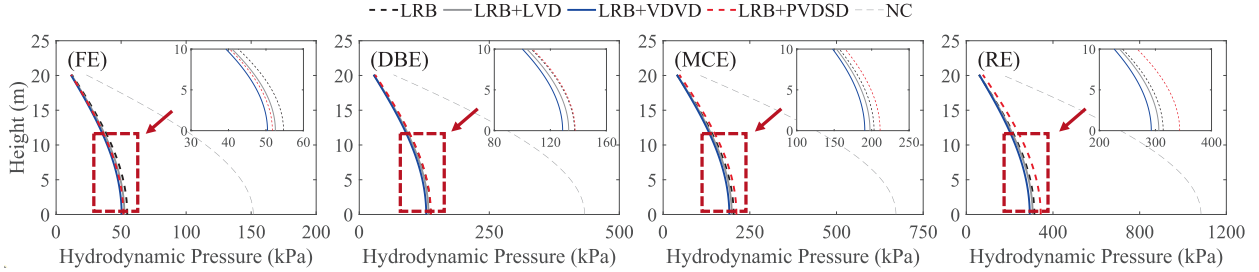
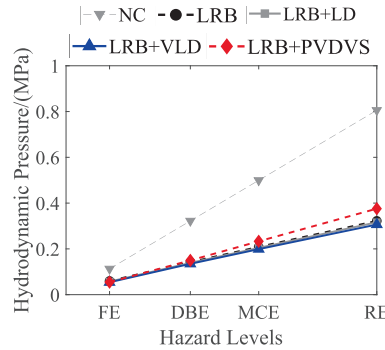
Fig. 26. Sloshing wave height responses under various seismic hazard levels: (a) Imperial Valley; (b) El Centro; (c) CHICHI.

Table 11

Average reduction rate of sloshing wave heights.

Response Indicators Hazard levels	Sloshing wave height FE	DBE	MCE	RE
$R_f/\%$ (Ito NC)	-2.1	12.4	19.8	27.3
$R_f/\%$ (IIto NC)	-2.1	11.3	17.2	23.9
$R_f/\%$ (IIIto NC)	-2.1	10.4	15.2	20.2
$R_f/\%$ (IVto NC)	-1.0	-7.2	-7.2	-8.7

Note: I: LRB; II: LRB + LVD; III: LRB + VDVS; IV: LRB + PVDSD.

**Fig. 27.** Hydrodynamic pressure along the height of the tank (Imperial Valley).**Fig. 28.** Peak hydrodynamic pressure under different hazard levels: (a) Imperial Valley; (b) El Centro; (c) CHICHI.

5. Conclusions

This study proposes a novel passive variable damping and stiffness device (PVDSD) and investigates its mechanical behavior and seismic mitigation performance for the Liquid Storage Tank (LST). A series of experiments were conducted to evaluate the variable damping and stiffness characteristics of the device. A nonlinear numerical model was developed and integrated into a base-isolated LST system. Subsequently, a comprehensive time-history analysis was carried out under a range of seismic hazard levels to assess the control effectiveness of the proposed LRB + PVDSD system in terms of isolation layer displacement, base shear, overturning moment, sloshing height and hydrodynamic pressure. The key conclusions are as follows:

- (1) The PVDSD device integrates variable damping and stiffness mechanisms for different excitation intensities. Performance tests show that the device can provide stable variable damping and spring forces. Under small displacement and low-frequency excitations, the device primarily dissipates energy through variable damping mechanism, with the variable stiffness mechanism remaining inactive. As the displacement increases, the stiffness components are progressively engaged, demonstrating a staged stiffness response that closely aligns with theoretical predictions.
- (2) A nonlinear numerical model of the PVDSD was established verified against test results. Then, it was implemented in a seismic isolation system combined with traditional lead-rubber bearings (LRBs). The proposed LRB + PVDSD system enables simultaneous control of damping and stiffness within the isolation layer, providing adaptive stiffness limitation under large displacements and enhanced damping under moderate excitations.
- (3) Time-history analyses under multiple earthquake records and hazard levels demonstrate the superior seismic performance of the LRB + PVDSD system. Compared with conventional LRB-based and other hybrid damping systems, the proposed system significantly reduces isolation layer displacement (with a maximum reduction rate exceeding 61 % under rare earthquakes),

while maintaining acceptable levels of base shear and overturning moment. Although a slight increase in sloshing wave height is observed due to stiffness activation, it does not adversely affect structural integrity. Overall, the LRB + PVDSD system achieves a well-balanced trade-off between displacement control and structural safety, making it a robust and practical isolation strategy for LST.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the writing process of this work, the authors used ChatGPT in order to improve language only. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Mingmei Shi: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Weiying Fu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis. **Mao Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Zheng Li:** Methodology, Formal analysis. **Haozhe Wang:** Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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